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§§ 108–116

Tenth Section.

Approximate computation of roots.

§ 108. Interpolation. Regula falsi

Once intervals have been separated in which only one root of an algebraic equation lies, it becomes possible to compute the real roots numerically to arbitrary accuracy by narrowing the interval more and more, for example by repeated bisection. If an interval of length 1 is repeatedly divided into ten parts, each new step supplies one further decimal place.

When a root has once been separated from all the others, then in the further narrowing of the interval one has only to take account of the sign of the function $f(x)$ itself. The computations needed for this may be greatly eased by applying the interpolation formula that we learned in the first section.

In § 10, (7), we derived a formula by which a function of degree n , which we shall now denote by $\varphi(\xi)$, can be determined when $n + 1$ consecutive values

$$\varphi(0), \quad \varphi(1), \dots, \varphi(n)$$

are given. This formula is, if

$$B_\nu^{(\xi)} = \frac{\xi(\xi-1)\cdots(\xi-\nu+1)}{1\cdot 2\cdot 3\cdots\nu}, \quad \Delta_\xi = \varphi(\xi+1) - \varphi(\xi), \quad (1)$$

$$\Delta'_\xi = \Delta_{\xi+1} - \Delta_\xi, \quad \Delta_\xi^{(\mu-1)} = \Delta_{\xi+1}^{(\mu-2)} - \Delta_\xi^{(\mu-2)} \quad (2)$$

are put,

$$\varphi(\xi) = \varphi(0) + \Delta_0 B_1^{(\xi)} + \Delta'_0 B_2^{(\xi)} + \cdots + \Delta_0^{(n-1)} B_n^{(\xi)}. \quad (3)$$

The value of the formula lies in the fact that the sequence of differences

$$\Delta_0, \quad \Delta'_0, \quad \Delta''_0, \dots$$

in many cases gives numerical values that decrease so rapidly that one may be content with a few of the first terms of the series (3).

Let $f(x)$ be the function to be investigated, and let (α, β) be the interval in which one of the required roots lies. If m denotes any suitably chosen integer, put

$$\delta = \frac{\beta - \alpha}{m}, \quad x = \alpha + \delta\xi,$$

and hence

$$\Delta_x = f(x + \delta) - f(x), \quad \Delta'_x = \Delta_{x+\delta} - \Delta_x, \quad \dots$$

Then

$$f(x) = f(\alpha) + \frac{\Delta_\alpha}{\delta}(x - \alpha) + \frac{\Delta'_\alpha}{\delta^2} \frac{(x - \alpha)(x - \alpha - \delta)}{2} + \cdots \quad (4)$$

One could just as well start from the other end point β of the interval, and would then have to put in (3)

$$x = \beta - \delta\xi.$$

Then one would get

$$\Delta_x = f(x - \delta) - f(x), \quad \Delta'_x = \Delta_{x-\delta} - \Delta_x, \quad \dots$$

and

$$f(x) = f(\beta) + \frac{\Delta_\beta}{\delta}(x - \beta) + \frac{\Delta'_\beta}{\delta^2} \frac{(x - \beta)(x - \beta + \delta)}{2} + \dots \quad (5)$$

If the equation

$$y = f(x)$$

is interpreted as the equation of a curve in rectangular coordinates x, y , then the relations may be illustrated geometrically. If, for example, equation (4) is applied to the interval from α to $\alpha + \delta$, and only the first difference is used, then in geometrical language the arc of the curve between the two points $\alpha, f(\alpha)$ and $\alpha + \delta, f(\alpha + \delta)$ is replaced by the chord. If the second difference is also taken into account, then the arc through the three consecutive points with abscissas $\alpha, \alpha + \delta, \alpha + 2\delta$ is replaced by the arc of a parabola passing through the same points and having its axis parallel to the ordinate axis; this parabola will adhere to the curve more closely than the chord.

The smaller the interval δ is, the less influence the higher differences have. How far one must go in the approximation depends not only on the accuracy with which one wishes to know $f(x)$, but essentially also on the density of the values

$$\alpha, \quad \alpha + \delta, \quad \alpha + 2\delta, \dots$$

for which the function is known. The arrangement of numerical tables, especially logarithm tables, rests on these propositions. Strictly speaking, the functions involved there are not integral rational functions; but for continuous functions generally the same laws hold here. Thus in the tables, beside the values

$$f(\alpha), \quad f(\alpha + \delta), \quad f(\alpha + 2\delta), \dots$$

one also finds the first or the first two differences. In the customary seven-place tables the first difference suffices. In Vega's ten-place table, the *Thesaurus logarithmorum*, the second differences are also printed and must be used in quite precise computations.

Our interpolation formulae may be used with advantage to compute the roots of equations, or, more exactly, to improve approximate values found by other means. We may formulate the problem as follows: for a given value of $f(x)$ between $f(\alpha)$ and $f(\alpha + \delta)$, find the corresponding value of x .

We regard $x = \alpha$ as a first approximation. Put

$$f(x) - f(\alpha) = \Delta.$$

Then Δ is a given value of the same sign as

$$\Delta_\alpha = f(\alpha + \delta) - f(\alpha)$$

and absolutely smaller than Δ_α . If further $x - \alpha = u$, formula (4) gives

$$\Delta = \frac{\Delta_\alpha}{\delta} u + \frac{\Delta'_\alpha}{\delta^2} \frac{u(u - \delta)}{2} + \dots \quad (6)$$

If one first stops with the first difference, the first correction is

$$u = \delta \frac{\Delta}{\Delta_\alpha}. \quad (7)$$

If now

$$u = \delta \frac{\Delta}{\Delta_\alpha} + u'$$

is put, then from (6), omitting u' in the second term, follows

$$0 = \frac{\Delta_\alpha}{\delta} u' + \frac{\Delta'_\alpha \Delta (\Delta - \Delta_\alpha)}{2\Delta_\alpha^2},$$

and hence as the second correction

$$u' = \delta \frac{\Delta'_\alpha \Delta (\Delta_\alpha - \Delta)}{2\Delta_\alpha^3}. \quad (8)$$

The first correction is obtained by replacing the arc between α and $\alpha + \delta$ by the chord; the second by replacing the arc through $\alpha, \alpha + \delta, \alpha + 2\delta$ by a parabola through the same points, but now with its axis parallel to the x -axis.

As an example take the equation

$$f(x) = x^3 - 2x - 2 = 0,$$

which has a real root between 1.7 and 1.8. One computes

x	$f(x)$	Δ_x	Δ'_x
1.7	−0.487	0.719	0.108
1.8	+0.232	0.827	
1.9	1.059		

Since $f(x) = 0$, one must put

$$\Delta = 0.487.$$

The first correction to $\alpha = 1.7$ is, by (7),

$$u = 0.06773,$$

and the second correction by (8) is

$$u' = 0.00164,$$

so that

$$\alpha + u + u' = 1.76937.$$

The more exact value computed in another way is 1.76929... Thus we have a result correct in the first three decimals. Here, however, we have no other means of estimating the accuracy beforehand than the decrease of the differences $\Delta, \Delta', \Delta'', \dots$. If Δ' is so small as to fall outside the limits of the intended accuracy, the use of the first difference gives an exact result for that purpose.

The rule for computing roots just explained is called the *regula falsi*. Its simplest form, suited for the first approximation, is this. If a root x of the equation $f(x) = 0$ lies between α and β , and if $f(\alpha) = -a$, $f(\beta) = b$, then

$$x = \frac{b\alpha + a\beta}{a + b} \quad (9)$$

is an approximate value of x . This is only another way of writing formula (7).

§ 109. Newton's method of approximation

A method that is usually better than interpolation for the approximate computation of roots of an equation goes back to Newton, and was developed and investigated more precisely by Fourier. The method is simply as follows. Let

$$f(x) = 0 \quad (1)$$

be the equation to be solved, and suppose that a value $x = \alpha$ has been found which may be regarded as a certain approximation to a root. Put in (1)

$$x = \alpha + h.$$

Then

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots . \quad (2)$$

If h is now determined from

$$f(\alpha) + hf'(\alpha) = 0, \quad (3)$$

which presupposes that $f'(\alpha)$ is not zero, then

$$f(\alpha + h) = \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots ,$$

and if h is small this value will be small, since only the square and higher powers of h occur. Thus, under suitable hypotheses,

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)} \quad (4)$$

may be regarded as a better approximation to the true value of the root. Replacing α by α' , one obtains

$$\alpha'' = \alpha' - \frac{f(\alpha')}{f'(\alpha')}$$

as a still better approximation, and so on.

Two questions remain to be answered:

1. Under what hypotheses is α' really a better value than α ?
2. How can one estimate the degree of accuracy that has thereby been reached?

Fourier answered these questions, but he makes the following assumption.

A root of $f(x)$ is enclosed in an interval (α, β) which contains no second root. In the interval (α, β) neither $f'(x)$ nor $f''(x)$ vanishes.

The latter assumption, that $f''(x)$ does not vanish in the interval, is made only in order to obtain conditions for applying the method that are easier to express. The usefulness of the method itself, once the interval is sufficiently narrowed, does not depend on it. If $f(x)$ and $f''(x)$ have no common divisor, so that they do not vanish simultaneously, then Fourier's assumption can always be satisfied by narrowing the interval; and if $f(x)$ and $f''(x)$ do have a common divisor, one may first separate it and then apply the approximation method to the individual factors of $f(x)$.

The relations are most easily surveyed geometrically, by interpreting $y = f(x)$ as the equation of a plane curve in rectangular coordinates. The equation

$$y = f(\alpha) + (x - \alpha)f'(\alpha) \quad (3)$$

is the equation of the tangent to the curve at the point $\alpha, f(\alpha)$; thus Newton's method amounts to replacing the curve in the interval (α, β) in first approximation by the tangent at one end point, rather than by its chord as in the interpolation method.

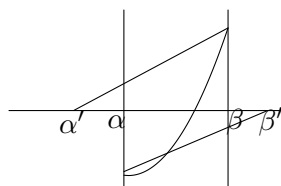


Fig. 21.

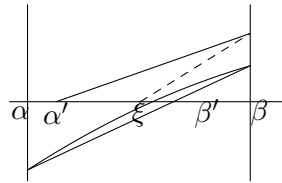


Fig. 22.

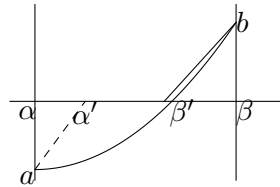


Fig. 23.

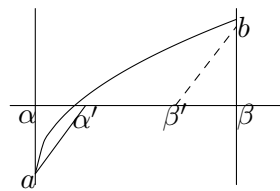


Fig. 24.

If the second differential quotient vanishes in the interval (α, β) , so that the curve has a point of inflection, it may happen that both end tangents lead out of the interval; then Newton's method is not applicable (Fig. 21). But it may also happen, when the interval (α, β) has already been sufficiently narrowed, that both tangents lead to interior points of the interval (Fig. 22). In these cases, however, the regula falsi will always give a better approximation.

We now assume that $f'(x)$ and $f''(x)$ do not vanish in (α, β) . Then the curve does not change the direction of its curvature, and certainly one of the two end tangents has its intersection with the x -axis inside the interval. This is certainly the case if the tangent is taken at the end point of the interval at which $f(x)$ and $f''(x)$ have the same sign. Figures 23 and 24 illustrate the relation for positive $f'(x)$ and positive, respectively negative, $f''(x)$. Thus in the first case β' , and in the second α' , is a better approximate value than β or α .

If one wishes at the same time to move the other boundary in order to get a new narrower interval, then, following Fourier, one can draw from the other end point the line parallel to $b\beta'$; in the case of Fig. 23 this is done at α , and one obtains the point α' as the lower boundary of the new interval. In the first case one has

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\beta)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

and in the second case

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\alpha)}.$$

The interval (α', β') is narrower and contains the required root. It is hardly necessary to consider separately the two other cases in which $f'(x)$ is negative in the interval.

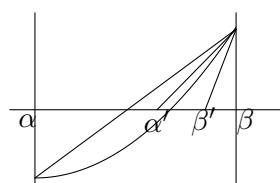


Fig. 25.

To obtain two new limits one may, with still better success, combine Newton's method with interpolation, as Fig. 25 shows. One then obtains the two new limits

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)} = \frac{\alpha f(\beta) - \beta f(\alpha)}{f(\beta) - f(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)}.$$

We summarize the result in the following proposition.

If an interval (α, β) is bounded in which $f(x)$ changes sign once, while $f'(x)$ and $f''(x)$ do not change sign, and if β is the end point at which $f(\beta)$ and $f''(\beta)$ have the same sign, whether β is smaller or greater than α , then a narrower interval (α', β') with the same properties is obtained by putting

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)}. \quad (4)$$

To prove these geometrically evident results analytically, suppose that in the interval $f'(x)$ and $f''(x)$ are positive, so that $\alpha < \beta$, and

$$f(\alpha) < 0, \quad f(\beta) > 0.$$

It is enough to treat this one case, since the three others are handled in exactly the same way.

Form the function

$$\varphi(x) = \frac{f(\beta) - f(x)}{\beta - x}$$

and its first derivative

$$\varphi'(x) = \frac{-(\beta - x)f'(x) + f(\beta) - f(x)}{(\beta - x)^2}.$$

The numerator

$$\psi(x) = -(\beta - x)f'(x) + f(\beta) - f(x)$$

vanishes for $x = \beta$, and its derivative is

$$\psi'(x) = -(\beta - x)f''(x),$$

hence is negative in the interval. Therefore $\psi(x)$ decreases as x increases and, since it vanishes for the greatest value $x = \beta$, remains positive. Consequently $\varphi'(x)$ is positive, and $\varphi(x)$ increases throughout the interval. Hence

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} < \frac{f(\beta) - f(x)}{\beta - x} < f'(\beta), \quad \alpha < x < \beta. \quad (5)$$

With

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

one has

$$\beta' - \alpha' = f(\beta) \left\{ \frac{\beta - \alpha}{f(\beta) - f(\alpha)} - \frac{1}{f'(\beta)} \right\}, \quad (6)$$

and by (5) this difference is positive. Thus

$$\alpha < \alpha' < \beta' < \beta.$$

Putting $x = \alpha'$ and $x = \beta'$ in (5), and observing that

$$\beta - \alpha' = \frac{(\beta - \alpha)f(\beta)}{f(\beta) - f(\alpha)}, \quad \beta - \beta' = \frac{f(\beta)}{f'(\beta)},$$

one gets

$$f(\alpha') < 0, \quad f(\beta') > 0.$$

Thus the required root is contained between α' and β' , and the interval (α', β') is smaller than (α, β) . Repeating the construction with α', β' in place of α, β , one obtains a new still narrower interval, and so on.

For convergence put, by (6),

$$\frac{\beta' - \alpha'}{\beta - \alpha} = \Phi(\alpha, \beta) = \frac{f(\beta)\{f(\alpha) - f(\beta) + (\beta - \alpha)f'(\beta)\}}{(\beta - \alpha)f'(\beta)\{f(\beta) - f(\alpha)\}}. \quad (7)$$

The numerator and denominator are both divisible by $(\beta - \alpha)^2$; after cancelling this common factor and writing ξ, η for α, β , one obtains an expression

$$\Phi(\xi, \eta) = \frac{\psi_1(\xi, \eta)}{\psi_2(\xi, \eta)},$$

where ψ_1 and ψ_2 are integral rational functions of the two variables ξ, η .

Let x be the root lying in (α, β) . Then, by our hypotheses on $f(x)$, the functions ψ_1 and ψ_2 do not vanish when

$$\alpha \leq \xi < x, \quad x < \eta \leq \beta. \quad (8)$$

The function $\Phi(\xi, \eta)$ is smaller than 1 for $\xi = \alpha, \eta = \beta$, and likewise for every other interval contained in (α, β) in which $f(\xi)$ is negative and $f(\eta)$ positive, since every such interval could have been used in place of (α, β) . Since $\Phi(\xi, \eta)$ is continuous in the two variables as long as they remain in the region (8), it must have there a maximum value, and this must be less than 1, because $\Phi(\xi, \eta)$ remains less than 1 also in the limiting cases $\xi = x, \eta = x$. Hence there is a positive proper fraction Θ such that

$$\Phi(\xi, \eta) < \Theta$$

throughout the region (8). It follows from (7) that

$$\beta' - \alpha' < (\beta - \alpha)\Theta.$$

The same reasoning applied to the step from (α', β') to (α'', β'') gives

$$\beta'' - \alpha'' < (\beta' - \alpha')\Theta',$$

where Θ' has for α', β' the same meaning as Θ for α, β . Since α', β' are in the region (8), Θ' is not greater than Θ , and Θ may be used in its place. Thus

$$\beta'' - \alpha'' < (\beta - \alpha)\Theta^2,$$

and in general

$$\beta^{(\nu)} - \alpha^{(\nu)} < (\beta - \alpha)\Theta^\nu.$$

The intervals therefore decrease at least as fast as the terms of a decreasing geometric progression.

As an example take

$$x^3 - 2x^2 - 2 = 0,$$

which has a root between

$$\alpha = 2.35, \quad \beta = 2.36.$$

From the formulae (4) one obtains

$$\beta' = 2.35931\dots, \quad \alpha' = 2.359298\dots,$$

so that a value correct in the fourth decimal is

$$2.3593.$$

For this value itself, a more exact computation shows that $f(x)$ is still negative; it can therefore be taken as α' . The next approximation gives

$$2.359304.$$

§ 110. The approximation method of Daniel Bernoulli and related methods

The method for approximately solving an equation that goes back to Daniel Bernoulli rests on the fact that, if one has a sequence of real quantities, the powers of the greatest among them dominate the equal powers of the others more and more as the exponents grow.

Let $\alpha, \beta, \gamma, \dots$ be arbitrary real or complex quantities, but such that the absolute value of α is greater than the absolute value of all the others; that is, the absolute values of $\beta : \alpha, \gamma : \alpha, \dots$ are proper fractions. Then

$$\frac{\alpha^m + \beta^m + \gamma^m + \dots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \dots} = \alpha \frac{1 + \left(\frac{\beta}{\alpha}\right)^m + \left(\frac{\gamma}{\alpha}\right)^m + \dots}{1 + \left(\frac{\beta}{\alpha}\right)^{m-1} + \left(\frac{\gamma}{\alpha}\right)^{m-1} + \dots}. \quad (1)$$

As m increases, this expression approaches the limit α . If $\alpha, \beta, \gamma, \dots$ are the roots of an algebraic equation, the left hand side of (1) is the quotient of the m th and $(m-1)$ st power sums, and therefore:

The quotient of the m th and $(m-1)$ st power sums approaches, as m increases, the root that is absolutely greatest.

Taking m negative and supposing α absolutely smaller than $\beta, \gamma, \dots$, one similarly obtains:

The quotient of the $-m$ th and $-(m+1)$ st power sums approaches, as m increases, the root that is absolutely smallest.

Since power sums can be computed from the coefficients as symmetric functions, one need only take m sufficiently large to obtain an approximate value of the absolutely greatest or absolutely smallest root. If among the roots of the equation there are roots with the same absolute value, and this value is the greatest or the smallest, then the method is not applicable; thus, for instance, it is not applicable to real equations in which a pair of imaginary roots has greater absolute value than the real roots. If the greatest root occurs singly, but exceeds the next only slightly, the process gives accuracy only slowly.

Jacobi supplemented Bernoulli's method in one direction.¹ Suppose that among the roots $x_1, x_2, \dots, x_n$ the roots

$$x_1, x_2, \dots, x_k$$

have absolute values greater than those of $x_{k+1}, x_{k+2}, \dots, x_n$, and put

$$\begin{aligned} p_m &= x_1^m + x_2^m + \dots + x_k^m, \\ p_{m+1} &= x_1^{m+1} + x_2^{m+1} + \dots + x_k^{m+1}, \\ &\vdots \\ p_{m+k} &= x_1^{m+k} + x_2^{m+k} + \dots + x_k^{m+k}. \end{aligned} \quad (2)$$

If

$$(x - x_1)(x - x_2) \dots (x - x_k) = x^k + A_1 x^{k-1} + \dots + A_k, \quad (3)$$

then the coefficients $A_1, A_2, \dots, A_k$ satisfy

$$\begin{aligned} p_{m+k} + p_{m+k-1}A_1 + p_{m+k-2}A_2 + \dots + p_m A_k &= 0, \\ p_{m+k+1} + p_{m+k}A_1 + p_{m+k-1}A_2 + \dots + p_{m+1}A_k &= 0, \\ &\vdots \\ p_{m+2k-1} + p_{m+2k-2}A_1 + p_{m+2k-3}A_2 + \dots + p_{m+k-1}A_k &= 0. \end{aligned} \quad (4)$$

¹Observatiunculæ etc., Works, vol. 3, p. 280.

If the equation

$$x^k + A_1x^{k-1} + A_2x^{k-2} + \cdots + A_k = 0$$

is adjoined to (4) and the A_i are eliminated, one obtains the equation of degree k

$$\begin{vmatrix} x^k & x^{k-1} & \cdots & 1 \\ p_{m+k} & p_{m+k-1} & \cdots & p_m \\ p_{m+k+1} & p_{m+k} & \cdots & p_{m+1} \\ \vdots & \vdots & & \vdots \\ p_{m+2k-1} & p_{m+2k-2} & \cdots & p_{m+k-1} \end{vmatrix} = 0,$$

whose roots are $x_1, x_2, \dots, x_k$. The quantities $p_m, p_{m+1}, \dots$ may, with increasing accuracy, be replaced by the corresponding power sums of all the roots as m increases.

For example, when $k = 2$ one gets

$$x^2 + \frac{p_m p_{m+3} - p_{m+1} p_{m+2}}{p_{m+1}^2 - p_m p_{m+2}} x + \frac{p_{m+2}^2 - p_{m+1} p_{m+3}}{p_{m+1}^2 - p_m p_{m+2}} = 0,$$

an equation applicable when, in a real equation, a pair of conjugate roots has the greatest absolute value.

Applying Bernoulli's method to the sum of negative powers gives, as observed, an approximation to the absolutely smallest root. But by shifting the origin one can make any root the absolutely smallest one, although this requires knowledge of a value already approximated to a certain degree.

A formula sufficient for all cases was given by Fr. Meyer.² Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of $f(x) = 0$, real or imaginary. Seek a point p in the x -plane such that a circle with centre p contains only one of the points $\alpha_1, \alpha_2, \dots, \alpha_n$, say α_1 . Then the absolute values of

$$\frac{p - \alpha_1}{p - \alpha_2}, \quad \frac{p - \alpha_1}{p - \alpha_3}, \dots, \frac{p - \alpha_1}{p - \alpha_n} \quad (5)$$

are proper fractions. Such points always exist; if necessary they must be found by Sturm's method. Choose also any function $\varphi(x)$ that has no common divisor with $f(x)$, for instance $\varphi(x) = 1$, and form the symmetric functions

$$y_m = \frac{\frac{\alpha_1 \varphi(\alpha_1)}{(p - \alpha_1)^m} + \frac{\alpha_2 \varphi(\alpha_2)}{(p - \alpha_2)^m} + \cdots + \frac{\alpha_n \varphi(\alpha_n)}{(p - \alpha_n)^m}}{\frac{\varphi(\alpha_1)}{(p - \alpha_1)^m} + \frac{\varphi(\alpha_2)}{(p - \alpha_2)^m} + \cdots + \frac{\varphi(\alpha_n)}{(p - \alpha_n)^m}}. \quad (6)$$

Then y_m tends to α_1 as m increases, and the convergence is faster the smaller the absolute values of the fractions (5) already are. This is immediate if (6) is written as

$$y_m = \frac{\alpha_1 \varphi(\alpha_1) + \alpha_2 \varphi(\alpha_2) \left(\frac{p - \alpha_1}{p - \alpha_2} \right)^m + \cdots + \alpha_n \varphi(\alpha_n) \left(\frac{p - \alpha_1}{p - \alpha_n} \right)^m}{\varphi(\alpha_1) + \varphi(\alpha_2) \left(\frac{p - \alpha_1}{p - \alpha_2} \right)^m + \cdots + \varphi(\alpha_n) \left(\frac{p - \alpha_1}{p - \alpha_n} \right)^m}.$$

§ 111. Gräffe's method of approximation

A procedure resting on a similar idea to Bernoulli's approximation was given by Gräffe and further developed by Encke.³ It is especially suited to a practical execution of numerical computations.

Let $\alpha, \beta, \gamma, \dots$ be the roots of an equation, and let α be absolutely the greatest of them, no other root being equal to α in absolute value. Then

$$\sqrt[m]{\alpha^m + \beta^m + \gamma^m + \cdots} \quad (1)$$

²Mathematische Annalen, vol. 33.

³Crelle's Journal, vol. 22 (1841).

has the limit α as m grows without bound; indeed its convergence to α is even better than for the quotient considered in the previous paragraph,

$$\frac{\alpha^m + \beta^m + \gamma^m + \dots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \dots}.$$

This is seen by putting, by the generalized binomial theorem,

$$\begin{aligned} \sqrt[m]{\alpha^m + \beta^m + \gamma^m + \dots} &= \alpha \sqrt[m]{1 + \left(\frac{\beta}{\alpha}\right)^m + \left(\frac{\gamma}{\alpha}\right)^m + \dots} \\ &= \alpha \left[1 + \frac{1}{m} \left(\frac{\beta}{\alpha}\right)^m + \frac{1}{m} \left(\frac{\gamma}{\alpha}\right)^m + \dots \right]. \end{aligned}$$

The absolutely smallest root is obtained in the same way as the limit of

$$\frac{1}{\sqrt[m]{\frac{1}{\alpha^m} + \frac{1}{\beta^m} + \frac{1}{\gamma^m} + \dots}}, \quad (2)$$

and, if p is arbitrary, the root nearest to p is the limit of

$$p + \frac{1}{\sqrt[m]{\frac{1}{(\alpha-p)^m} + \frac{1}{(\beta-p)^m} + \dots}}. \quad (3)$$

By transforming the equation $f(x)$ by the substitutions

$$y = \frac{1}{x}, \quad y = \frac{1}{x-p},$$

the expressions (2) and (3) are obtained from (1), to which we therefore restrict ourselves.

Applying (1) to a real equation, $\alpha^m + \beta^m + \gamma^m + \dots$ is a real number; and if the absolutely greatest root α is real, the m th root is to be taken real. If m is even, one must also know, to decide the sign, whether α is positive or negative. If necessary this is decided by substituting the approximated value in the given equation. Likewise, if (1) is applied to an imaginary equation, one must decide which of the several m th roots gives the correct approximation.

The quantities

$$s_m = \alpha^m + \beta^m + \gamma^m + \dots$$

are particularly easy to compute when m is taken to be successive powers of 2, that is $m = 2, 4, 8, 16, \dots$, and then give very good results. The number s_2 is the negative coefficient of the $(n-1)$ st power of the unknown in the equation whose roots are the squares of the roots of $f(x)$. This equation is easily formed as follows. Collect the terms of $f(x)$ with even and odd powers of x and put

$$f(x) = \varphi(x^2) + x\psi(x^2).$$

Then

$$f_1(x) = \varphi(x)^2 - x\psi(x)^2 = 0$$

is the equation whose roots are the squares of the roots of $f(x)$. Treating $f_1(x)$ in the same manner gives an equation whose roots are the fourth powers of the roots of $f(x)$, and so on. The computation is very simple and after few steps usually leads to a good approximation.

First consider

$$x^3 - 2x - 2 = 0, \quad (4)$$

which has one positive real and two imaginary roots. The real root is greater than 1.7, and since the product of all three roots is 2 and $(1.7)^3 > 2$, the absolute value of each of the two imaginary

roots is smaller than the real root. Thus the real root is the absolutely greatest one and Gräffe's method must lead to it. The equations whose roots are the second, fourth, eighth, ... powers of the roots of (4) are

$$x^3 - 4x^2 + 4x - 4 = 0,$$

$$x^3 - 8x^2 - 16x - 16 = 0,$$

$$x^3 - 96x^2 - 16^2 = 0,$$

and

$$x = \sqrt[8]{96} = 1.7692 \dots$$

is already correct in the first four decimals. The next step gives no new result, because the first power of the unknown is missing in the last equation; but the following step gives the value

$$x = \sqrt[32]{85032960} = 1.769293 \dots,$$

correct in the sixth decimal.

As a second example take a fifth-degree equation, which also gives occasion to apply several propositions of the preceding paragraph. As before, the example is taken from the theory of complex multiplication of elliptic functions. The equation is

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0. \quad (5)$$

We derive the equation whose roots are the squares of the roots of (5) by replacing x^2 by x in

$$(x^5 - x^3 - 2x)^2 - (2x^2 + 1)^2 = 0.$$

This gives

$$x^5 - 2x^4 - 3x^3 - 1 = 0, \quad (6)$$

and applying the same procedure twice more yields the equations whose roots are the fourth and eighth powers of the roots of (5):

$$x^5 - 10x^4 + 9x^3 - 4x^2 - 1 = 0, \quad (7)$$

$$x^5 - 82x^4 + x^3 - 36x^2 - 8x - 1 = 0. \quad (8)$$

The last of these equations is suited to discussion. By Descartes' theorem (§ 101) it has at least one positive root. If, for equation (8), one forms the functions that occur in Newton's criterion (§ 101),

$$\begin{array}{ccccc} 1, & 4a_0^2 - 10a_0a_2, & 6a_2^2 - 12a_1a_3, \\ 6a_2^2 - 12a_3a_1, & 4a_4^2 - 10a_3a_5, & 1, \end{array}$$

one obtains the signs

$$+ \quad + \quad - \quad + \quad - \quad +$$

and (8) therefore has, by § 101, theorem VIII, four imaginary roots. The positive root is certainly greater than 1, since the left side of (8) is negative for $x = 0$ and $x = 1$.

To decide which root has the greatest absolute value, put $x = 1 : z$ in (8) and then, following § 96, 97, investigate how many roots lie inside the unit circle in the z -plane. Thus in

$$z^5 + 8z^4 + 36z^3 - z^2 + 82z - 1$$

put

$$z = \cos \vartheta + i \sin \vartheta.$$

The real and imaginary parts are

$$\varphi = \cos 5\vartheta + 8 \cos 4\vartheta + 36 \cos 3\vartheta - \cos 2\vartheta + 82 \cos \vartheta - 1,$$

$$\psi = \sin 5\vartheta + 8 \sin 4\vartheta + 36 \sin 3\vartheta - \sin 2\vartheta + 82 \sin \vartheta.$$

It can be shown that ψ is positive when $0 < \vartheta < \pi$ and negative when $0 > \vartheta > -\pi$, so that the unit circle is divided into two segments in which ψ has opposite signs. It follows by §96 that only one root can lie inside the circle, namely the real root, while the imaginary roots lie outside. Hence, for equation (8), the real root has the greatest absolute value and is therefore found by Gräffe's method.

To prove this property of ψ , put $2 \cos \vartheta = \xi$ and use the trigonometric formulae

$$\begin{aligned}\sin 5\vartheta &= \sin \vartheta(\xi^4 - 3\xi^2 + 1), \\ \sin 4\vartheta &= \sin \vartheta(\xi^3 - 2\xi), \\ \sin 3\vartheta &= \sin \vartheta(\xi^2 - 1), \\ \sin 2\vartheta &= \sin \vartheta \xi.\end{aligned}$$

This gives

$$\psi = \sin \vartheta \{\xi^4 + \xi^2(8\xi + 33) + (47 - 17\xi)\},$$

from which it is immediately seen that the factor of $\sin \vartheta$ remains positive as long as ξ lies between -2 and $+2$.

Equation (8) itself already gives a fairly good approximation to the real root,

$$x = \sqrt[8]{82} = 1.73471.$$

A more exact value, obtained by going two more steps, is 1.73469, still correct in the fifth decimal.

This method does not enclose the required root between two bounds. The sign that a sufficient degree of accuracy has been reached is that, if s_μ and $s_{\mu'}$ are two successive power sums of the roots used for the computation, then they behave with sufficient accuracy as the μ th and μ' th powers of a quantity; equivalently, approximately

$$\mu \log s_{\mu'} - \mu' \log s_\mu = 0.$$

One must carry out this check for several successive terms, because in special cases the relation can appear to be satisfied and then fail again later in the computation.

§ 112. Trigonometric solution of cubic equations

We next discuss a few methods of numerical solution that apply to equations of special forms. Their aim is to make the widely available tables of trigonometric functions and logarithms useful for solving equations. We first turn to cubic equations, which we always assume in the reduced form

$$x^3 + ax + b = 0, \tag{1}$$

where a and b are real numbers. Replacing x by $-x$ amounts to replacing b by $-b$, so we may assume that b is negative. There remain three cases. Put $b = -g$, and according as a is positive or negative put $a = \pm e$. Then

$$\begin{aligned}\text{a)} \quad & x^3 + ex - g = 0, \\ \text{b)} \quad & x^3 - ex - g = 0, \quad \frac{e^3}{27} < \frac{g^2}{4}, \\ \text{c)} \quad & x^3 - ex - g = 0, \quad \frac{e^3}{27} > \frac{g^2}{4}.\end{aligned}$$

The limiting cases in which one of $e, g, 4e^3 - 27g^2$ vanishes are excluded.

To apply Cardano's formula (§35), set

$$R = \frac{g^2}{4} \pm \frac{e^3}{27}, \tag{2}$$

where the upper sign applies in case a), and the lower in cases b) and c). In cases a) and b) only one root is real; in case c), the irreducible case, all three roots are real. Cardano's formula gives

$$x = \sqrt[3]{\frac{g}{2} + \sqrt{R}} + \sqrt[3]{\frac{g}{2} - \sqrt{R}}. \quad (3)$$

Proceed as follows in the three cases.

a) Introduce an angle ϑ , chosen between 0° and 90° , by

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}} \cotg \vartheta, \quad (4)$$

so that

$$\sqrt{R} = \sqrt{\frac{e^3}{27}} \frac{1}{\sin \vartheta},$$

and

$$x = \sqrt{\frac{e}{3}} \left(\sqrt[3]{\cotg \vartheta + \frac{1}{\sin \vartheta}} + \sqrt[3]{\cotg \vartheta - \frac{1}{\sin \vartheta}} \right) = \sqrt{\frac{e}{3}} \left(\sqrt[3]{\cotg \frac{\vartheta}{2}} - \sqrt[3]{\tg \frac{\vartheta}{2}} \right).$$

If an angle φ is now determined from

$$\tg \frac{\vartheta}{2} = \tg^3 \varphi, \quad (5)$$

then

$$x = 2\sqrt{\frac{e}{3}} \cotg 2\varphi. \quad (6)$$

The angles ϑ, φ and finally x are found from the logarithmic trigonometric tables by formulae (4), (5), (6). To obtain the imaginary roots, replace $\tg \varphi$ by $\varrho \tg \varphi$, where ϱ denotes an imaginary cube root of unity.

b) In the second case determine ϑ , again in the first quadrant, from

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}} \frac{1}{\sin \vartheta}, \quad (7)$$

$$\sqrt{R} = \sqrt{\frac{e^3}{27}} \cotg \vartheta, \quad x = \sqrt{\frac{e}{3}} \left(\sqrt[3]{\cotg \frac{\vartheta}{2}} + \sqrt[3]{\tg \frac{\vartheta}{2}} \right),$$

$$\tg \frac{\vartheta}{2} = \tg^3 \varphi, \quad (8)$$

$$x = 2\sqrt{\frac{e}{3}} \frac{1}{\sin 2\varphi}. \quad (9)$$

c) Finally, in the last case, where three real roots occur, put

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}} \cos \vartheta, \quad (10)$$

$$\sqrt{R} = i\sqrt{\frac{e^3}{27}} \sin \vartheta,$$

$$x = \sqrt{\frac{e}{3}} \left(\sqrt[3]{\cos \vartheta + i \sin \vartheta} + \sqrt[3]{\cos \vartheta - i \sin \vartheta} \right).$$

By Moivre's theorem,

$$x = 2\sqrt{\frac{e}{3}} \cos \frac{\vartheta}{3}. \quad (11)$$

Taking ϑ again in the first quadrant, the other two real roots are

$$-2\sqrt{\frac{e}{3}} \cos \frac{\pi + \vartheta}{3}, \quad -2\sqrt{\frac{e}{3}} \cos \frac{\pi - \vartheta}{3}.$$

All these formulae are arranged for logarithmic computation.

§ 113. Gauss's method for solving trinomial equations

Gauss gave a method for numerically solving, in a simple way, the roots of an equation containing only three terms. Such equations occur frequently and include, as special cases, all quadratic and all reduced cubic equations.

For such an equation, in the general form

$$x^{m+n} + ax^m + b = 0, \quad (1)$$

only the positive root, if it exists, is first sought. The possible negative root is obtained by replacing x by $-x$. According to the signs of a and b there are three cases, since if both signs are positive no positive root exists. Thus, with e and g positive,

$$\begin{aligned} \text{a)} \quad & x^{m+n} + ex^m - g = 0, \\ \text{b)} \quad & x^{m+n} - ex^m - g = 0, \\ \text{c)} \quad & x^{m+n} - ex^m + g = 0. \end{aligned}$$

In the first two cases Descartes' theorem gives one positive root; in the third there may be two positive roots or none.

Put, with Gauss,

$$\lambda = \frac{g^m}{e^{m+n}},$$

and reduce the three equations a), b), c) by suitable substitutions to

$$\sin^2 \Theta + \cos^2 \Theta = 1.$$

Use

$$\begin{aligned} \text{a)} \quad & \frac{x^{m+n}}{g} = \sin^2 \Theta, \quad \frac{ex^m}{g} = \cos^2 \Theta, \quad \lambda = \frac{\sin^{2m} \Theta}{\cos^{2m+2n} \Theta}, \\ \text{b)} \quad & gx^{-m-n} = \sin^2 \Theta, \quad ex^{-n} = \cos^2 \Theta, \quad \lambda = \frac{\sin^{2m} \Theta}{\cos^{2m+2n} \Theta}, \\ \text{c)} \quad & \frac{x^n}{e} = \sin^2 \Theta, \quad \frac{gx^{-m}}{e} = \cos^2 \Theta, \quad \lambda = \sin^{2m} \Theta \cos^{2n} \Theta. \end{aligned} \quad (2)$$

The last equation distinguishes the two cases in which equation c) has two positive roots or none. The maximum of $\sin^{2m} \Theta \cos^{2n} \Theta$ is

$$\frac{m^m n^n}{(m+n)^{m+n}},$$

and is reached when $\operatorname{tg} \Theta^2 = m : n$. If λ is below this bound, case c) gives two real roots; otherwise none.

In (2), λ is known, and one has to find from the tables the angle Θ satisfying the equation. If one has as yet no knowledge of the approximate position of this angle, it is expedient for the first approximation to use a table proceeding degree by degree and rounded to two decimals, and then to improve the value thus found with more accurate tables.

Gauss does not use trigonometric tables, but the addition and subtraction logarithms first introduced by him. We briefly show this in one case; the details for practical application are to be sought in Gauss's paper.⁴

The tables of addition and subtraction logarithms, as first introduced and computed by Gauss and as now found in common table works, give, for three numbers greater than 1,

$$a, \quad b = 1 + \frac{1}{a}, \quad c = 1 + a,$$

⁴Beiträge zur Theorie der algebraischen Gleichungen, second part (1849). Gauss, Works, vol. III, p. 85.

the Briggsian logarithms

$$A, \quad B, \quad C.$$

If Θ lies in the first octant, one can put

$$a = \cotg \Theta^2, \quad b = \frac{1}{\cos \Theta^2}, \quad c = \frac{1}{\sin \Theta^2}, \quad 0 < \Theta < 45^\circ, \quad (3)$$

and if Θ lies in the second octant,

$$a = \tg \Theta^2, \quad b = \frac{1}{\sin \Theta^2}, \quad c = \frac{1}{\cos \Theta^2}, \quad 45^\circ < \Theta < 90^\circ. \quad (4)$$

Apply this to case b). One must distinguish whether λ is less or greater than 2^m , since that determines whether Θ lies in the first or second octant. Thus:

$$\begin{aligned} \alpha) \quad \lambda < 2^m, \quad \lambda &= \frac{b^m}{a^n} = \frac{c^m}{a^{m+n}} = \frac{b^{m+n}}{c^n}, \\ x^{m+n} &= gc, \quad x^n = eb, \quad x^m = \frac{ga}{e}; \\ \beta) \quad \lambda > 2^m, \quad \lambda &= a^{m+n}b^m = a^m c^m = \frac{c^{m+n}}{b^n}, \\ x^{m+n} &= gb, \quad x^n = ec, \quad x^m = \frac{g}{ea}. \end{aligned} \quad (5)$$

In case α) one would put

$$\log \lambda = mB - nA \quad (6)$$

and then seek the corresponding values of A and B from the table. Once these are found,

$$m \log x = A + \log g - \log e. \quad (7)$$

As an example, take

$$x^3 - 2x - 2 = 0,$$

so that $e = g = 2$, $m = 1$, $n = 2$, $\lambda = \frac{1}{2}$. Formula (6) gives

$$2A - B = 0.3010300, \quad (8)$$

and (7)

$$\log x = A. \quad (9)$$

For a first approximate value of A one uses a small excerpt from the table:

A	B
0	0.301
0.1	0.254
0.2	0.212
0.3	0.176

This gives:

A	$2A - B$	error
0.2	0.188	-0.113
0.3	0.424	+0.123

The interpolation method may then be applied by dividing the total error 0.236 between the two values of A in the ratio of the partial errors:

$$A = 0.2 + \frac{0.1}{0.236} 0.113 = 0.247 \dots$$

With the help of a seven-place table of Zach one obtains:

A	B	$2A - B$	error
0.247	0.1948581	0.2991419	-0.0018881
0.248	0.1944969	0.3015031	+0.0004731

and by another interpolation

$$A = 0.2477996,$$

or

$$x = 1.769292.$$

§ 114. Computation of the imaginary roots of a trinomial equation

Gauss also gave a method for computing the imaginary roots of a trinomial equation. We briefly discuss it here. We assume real coefficients, although the method also applies in the general case. Put the equation in the form

$$x^{m+n} - ex^m - g = 0, \quad (1)$$

without here restricting e and g to positive values. Let

$$x = re^{i\vartheta}.$$

Equating the imaginary part in (1) to zero gives

$$r^n = \frac{e \sin m\vartheta}{\sin(n+m)\vartheta}. \quad (2)$$

Two more equations of this kind are obtained by multiplying (1) by x^{-m} and by x^{-m-n} :

$$r^{n+m} = -\frac{g \sin m\vartheta}{\sin n\vartheta}, \quad (3)$$

$$r^m = -\frac{g \sin(n+m)\vartheta}{e \sin n\vartheta}. \quad (4)$$

Any one of these three equations follows from the other two. Eliminating r from two of them gives, with

$$\lambda = \frac{g^m}{e^{n+m}},$$

$$\lambda = (-1)^n \frac{\sin^n n\vartheta \sin^m m\vartheta}{\sin^{m+n}(n+m)\vartheta}. \quad (5)$$

Since only one of the two conjugate roots has to be computed, ϑ may be restricted to the first quadrant; under some circumstances r must then be taken negative.

We illustrate how the angle ϑ is obtained from (5) using trigonometric tables, and then the corresponding value of r from one of (2), (3), (4), by the example already used:

$$x^3 - 2x - 2 = 0.$$

Here equation (5) becomes

$$\frac{1}{2} = \frac{\sin^2 2\vartheta \sin \vartheta}{\sin^3 3\vartheta}, \quad (6)$$

or

$$3 \log \sin 3\vartheta - 2 \log \sin 2\vartheta - \log \sin \vartheta = 0.3010300. \quad (7)$$

This first shows that ϑ lies between 0 and 60°. After a few trials one finds that ϑ lies between 33° and 34°. If for these two values one computes

$$3 \log \sin 3\vartheta - 2 \log \sin 2\vartheta - \log \sin \vartheta - 0.3010300, \quad (8)$$

then to three decimals one first obtains

$$0.026, \quad -0.013,$$

and by interpolation

$$\vartheta = 33^\circ 40'.$$

One then computes (8) somewhat more accurately, say to five places, for angles near $33^\circ 40'$, progressing minute by minute. From the change of sign one finds that ϑ lies between

$$33^\circ 41' \quad \text{and} \quad 33^\circ 42'.$$

Carrying out the calculation to seven places for these two values gives, again by interpolation,

$$\vartheta = 33^\circ 41' 20.6''.$$

From (3) or (4) one sees that here r is negative, and r is obtained very simply from one of these equations. The Briggsian logarithms are

$$\begin{aligned} \log(-r) &= 0.0266148, \\ \log \cos \vartheta &= 9.9201547 - 10, \\ \log \sin \vartheta &= 9.7440458 - 10. \end{aligned}$$

Thus the two imaginary roots are

$$-0.884646 \pm 0.589740 i.$$

Eleventh Section.

Continued fractions.

§ 115. Transformation of rational fractions into continued fractions

If one natural number m is divided by another n according to the ordinary rules, one obtains a quotient and a remainder, positive and smaller than the divisor n . If a denotes the quotient and r the remainder, then

$$m = an + r. \tag{1}$$

All numbers $m, m', m'', \dots$ that give the same remainder are called congruent modulo n , and Gauss denotes this by

$$m \equiv m' \pmod{n}. \tag{2}$$

Such a formula is called a congruence.

Thus, if we consider the system

$$r = 0, 1, 2, \dots, n-1,$$

every arbitrary positive or negative integer m is congruent modulo n to one and only one of these numbers. The same property is possessed by every system

$$s = m_0, m_1, m_2, \dots, m_{n-1}$$

chosen so that all residues r occur among their residues, and each occurs only once. Then every integer m is congruent modulo n to one and only one of the numbers s . We call such a system of numbers a complete residue system modulo n . For odd n one such complete residue system is formed by the numbers

$$\varrho = -\frac{n-1}{2}, -\frac{n-3}{2}, \dots, -1, 0, 1, 2, \dots, \frac{n-1}{2}.$$

The numbers r are called the least, or least positive, residues; the numbers ϱ the absolutely least residues.

To continue the division indicated in (1), denote m, n, r by m, m_1, m_2 , and denote the quotient and remainder of the division of m_1 by m_2 by a_1, m_3 , and so on. Also put a_0 for a . This gives a sequence of equations

$$\begin{aligned} m &= a_0 m_1 + m_2, \\ m_1 &= a_1 m_2 + m_3, \\ m_2 &= a_2 m_3 + m_4, \\ &\vdots \\ m_{\nu-1} &= a_{\nu-1} m_{\nu} + m_{\nu+1}, \end{aligned} \tag{3}$$

which may be continued as long as no remainder vanishes, that is, as long as no division is exact. Since

$$m_1 > m_2 > m_3 > \cdots,$$

and since there is only a finite number of positive integers smaller than m_1 , the division must terminate after finitely many steps. If m_{ν} is divisible by $m_{\nu+1}$, then $m_{\nu+1}$ is the greatest common divisor of m and m_1 . For, by (3), $m_{\nu+1}$ divides all preceding m_{μ} , and conversely every common divisor of two adjacent m_{μ} also divides all following terms, hence also $m_{\nu+1}$.

If therefore m and m_1 have no common divisor other than 1 (that is, if they are relatively prime), then $m_{\nu+1} = 1$, and (3) may be written

$$\frac{m}{m_1} = a_0 + \frac{m_2}{m_1}, \quad \frac{m_1}{m_2} = a_1 + \frac{m_3}{m_2}, \dots, \quad \frac{m_{\nu-1}}{m_{\nu}} = a_{\nu-1} + \frac{1}{m_{\nu}}. \tag{4}$$

This gives the expansion of the fraction $m : m_1$, or $m : n$, into a continued fraction. Successively,

$$\frac{m}{n} = a_0 + \frac{1}{\frac{m_1}{m_2}}, \quad \frac{m}{n} = a_0 + \frac{1}{a_1 + \frac{1}{\frac{m_2}{m_3}}}, \dots \tag{5}$$

and finally

$$\frac{m}{n} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_{\nu-1} + \frac{1}{m_{\nu}}}}}}. \tag{6}$$

For more convenient printing we also write

$$\frac{m}{n} = \left(a_0, \frac{m_1}{m_2} \right), \quad \frac{m}{n} = \left(a_0, a_1, \frac{m_2}{m_3} \right), \quad \frac{m}{n} = (a_0, a_1, a_2, \dots, a_{\nu-1}, m_{\nu}). \tag{7}$$

The following is still to be noted. Every positive rational fraction can be put in the form $m : n$ with m and n relatively prime; hence every such fraction can be expanded into a continued fraction of the form (6) or (7). The same holds for negative rational fractions, if negative values are allowed for a_0 . The numbers $a_1, a_2, \dots, a_{\nu-1}, m_{\nu}$ are then still positive integers. These numbers are completely and unambiguously determined. Only one arbitrariness remains.

By the determination just made, the last partial denominator m_{ν} is greater than 1, since $m_{\nu+1}$ was assumed to be the first remainder equal to 1. Hence, if a_{ν} is a positive integer, we may put either

$$m_{\nu} = a_{\nu}$$

or

$$m_\nu = a_\nu + \frac{1}{1}$$

and thus obtain the two continued-fraction representations

$$\frac{m}{n} = (a_0, a_1, a_2, \dots, a_\nu) = (a_0, a_1, a_2, \dots, a_\nu, 1). \quad (8)$$

I. Every rational fraction can therefore be expanded into a continued fraction having, at will, either an even or an odd number of partial denominators.

The numbers

$$\frac{m_1}{m_2}, \quad \frac{m_2}{m_3}, \quad \frac{m_3}{m_4}, \dots$$

that occur in (5) are called the final quantities of the continued fraction; the numbers $a_0, a_1, a_2, \dots$ will be called the partial denominators, although a_0 is not actually a denominator.

If m is replaced by a number m' congruent to it modulo n , then the two fractions $m : n$ and $m' : n$ differ only by an integer; their continued-fraction expansions therefore differ only in their first terms a_0 .

§ 116. Continued-fraction expansion of irrational numbers

If x is a real irrational number, then there is always one and only one integer a_0 such that x lies between a_0 and $a_0 + 1$. The integer a_0 is positive if $x > 1$, zero if x is a positive proper fraction, and negative if x is negative. Thus put

$$x = a_0 + \frac{1}{x_1}.$$

Then x_1 is a positive improper fraction; so if a_1 is the integer immediately below x_1 , then a_1 is positive. Put

$$x_1 = a_1 + \frac{1}{x_2}, \quad x_2 = a_2 + \frac{1}{x_3}, \quad \dots, \quad x_{n-1} = a_{n-1} + \frac{1}{x_n}.$$

Thus $x_1, x_2, \dots, x_n$ are positive improper fractions, and $a_1, a_2, \dots, a_{n-1}$ are positive integers. In the notation of the preceding paragraph, x is therefore the continued fraction

$$(a_0, a_1, a_2, \dots, a_{n-1}, x_n).$$

If x is irrational, this continued-fraction expansion can be continued without bound; that is, n may be taken arbitrarily large. The following expansion arises from the preceding one by putting

$$x_n = a_n + \frac{1}{x_{n+1}}.$$

Here also the integers $a_0, a_1, a_2, \dots, a_{n-1}$ are called the partial denominators, and x_n the final quantity.

The continued-fraction expansion of rational numbers is contained in this as a special case. By proposition I, a continued fraction can be continued at most by one term when the final quantity is an integer.